

Moritz Gilges

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SUMMARY

Computer Science M.Sc. student at the University of Bonn specializing in machine learning, with hands-on software engineering experience as a working student at fulfillmenttools. Strong foundations in deep learning, data science and NLP, combined with production experience shipping code in Python, TypeScript and Java. Increasingly drawn to research, aiming to bridge rigorous ML methods with reliable, production-ready systems.

EDUCATION

Rheinische-Friedrich-Wilhelm University

M.Sc., Computer Science, Intelligent systems track

Bonn, Germany

Oct 2025 - Present

Heinrich-Heine University

B.Sc., Computer Science, 1.8

Düsseldorf, Germany

Oct 2021 - Sep 2025

Thesis (1.0): Building a Web-Scraper for Academic Research

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, C, SQL, HTML/CSS, Bash/Shell

ML / AI: PyTorch, HF, LangChain, LangGraph, CrewAI, Pydantic – RAG, LoRA, NLP, MCP, multi agent systems

Frameworks / Libraries: NumPy, Pandas, scikit-learn, Flask, Spring

Developer Tools: Git, Docker, GCP, ChromaDB, MongoDB, PostgreSQL, FAISS, Maven/Gradle, Selenium, pytest

EXPERIENCE

Working Student - Backend Engineer

fulfillmenttools

Jul. 2025 – Present

Cologne, Germany

- Develop and maintain features in the inventory domain of a multi-tenant order-fulfillment SaaS platform (TypeScript, MongoDB, GCP)
- Built a scheduled cron job that detects and safely purges zero-quantity stock records at scale (hundreds of thousands of entries across tenants)
- Implemented asynchronous work queues and scheduled jobs for reliable background operations on large datasets
- Designed and refactored fine-grained permission logic and extended REST and GraphQL APIs to support new product requirements
- Wrote automated Jest test suites and integrated them into CI/CD pipelines to safeguard inventory services in production

Working Student - Software Engineer

adesso, Digital Experience

Apr. 2023 - Dec. 2023

Düsseldorf, Germany

- Developed and maintained CMS-based websites and Java plugins for internal content-management systems
- Drove the migration of multiple internal projects from Maven to Gradle, establishing the new build standard
- Set up CI/CD pipelines and led internal trainings on the new build tooling

Internship - IT Support

Colegio Aleman

Aug. 2021 - Sep. 2021

Quito, Ecuador

- Provided monitoring and support for 100+ computers and network access points across the school
- Configured PCs, tablets, printers and interactive whiteboards (Wi-Fi, antivirus, Office, disk imaging) and resolved hands-on technical issues in classrooms

PROJECTS

- Biomedical Entity Disambiguation Pipeline** (b-it Lab) | *Python, LLMs, RAG* April 2026 – Present
- Built an end-to-end pipeline that resolves ambiguous biomedical terms to their correct database entries, reducing manual annotation effort for researchers
 - Implemented embedding-based candidate retrieval across multiple biomedical and general knowledge bases (MeSH, UMLS, Wikidata, DBpedia) using FAISS similarity search, combined with domain-specific filtering rules
 - Integrated a locally hosted LLM (via LM Studio) for final disambiguation, keeping all data on-premise so sensitive research data never leaves the infrastructure
 - Fine-tuned the model with LoRA on the university GPU cluster Bender; structured LLM outputs with Pydantic AI for reliable, schema-validated results
- Narrative Story Similarity** (*SemEval-2026 Shared Task*) | *Python, SBERT, spaCy* Oct. 2025 – Feb. 2026
- Compared prompt engineering (GPT-4o-mini, zero-/few-shot) against representation learning (Sentence-BERT) for narrative similarity across theme, plot and outcome, on a dataset of 2,500+ labeled story triplets
 - Built a spaCy preprocessing pipeline (tokenization, POS, dependency parsing, NER) with a pseudonymization variant that replaces named entities, exposing how strongly models rely on surface features
 - Identified and analyzed a key negative result, triplet-loss fine-tuning degraded embeddings to chance level, tracing it to a mismatch between synthetic supervision and the similarity objective; presented findings in a poster session

CERTIFICATIONS

RAG and Agentic AI - IBM

Issued Aug. 2026

Introduction to agent skills - Anthropic

Issued May 2026

Claude 101 - Anthropic

Issued May 2026